

Experiment 3: Automatic Classification and Arrangement of Desktop Objects

Group 3

Group Laboratory Report

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Robotic Systems Integration Laboratory

Elephant Robotics mechArm 270 Pi, Isaac Sim, ROS 2 Humble, and YOLO

Abstract

This group report combines the simulation and real-robot validation stages of Experiment 3. The simulation implements automatic classification and arrangement of tennis balls and green wooden pencils on a circular desktop. A simulated mechArm 270 Pi is placed at the circle centre, while six objects and two open-top bins are arranged on the same ring. YOLO determines the category at a known grid coordinate, while the pickup position comes from fixed world coordinates. The final workflow buffers six consecutive camera frames and applies a 4-out-of-6 majority vote before executing the G1–G6 pick-and-place sequence.

The simulation uses Isaac Sim 5.1.0 on Windows, ROS 2 Humble in Docker, and TCP bridges for joint commands and camera frames. A 13-view sweep tested 65 valid images. The selected 10-degree north-tilted view achieved 6/6 correct fixed-ROI classifications in every tested frame, with mean confidence 0.836. The real stage uses a RoboMaster EP with an arm-mounted camera and a direct Python SDK program. The representative real run completed two source-grid-clearing sorting actions, one bottle and one cup, and preserved scan images, task logs, and video evidence. The supplied real package reports 88 offline unit tests passed. The report distinguishes this two-object real demonstration from the six-object simulation acceptance boundary.

Keywords: robotic manipulation; Isaac Sim; ROS 2; fixed-point grasping; YOLO; numerical inverse kinematics

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1 Basic Requirements of the Experiment

The objective of Experiment 3 is to build a simulation in which a robotic arm identifies desktop objects by category and places them into corresponding containers. The required categories in this implementation are `tennis_ball` and `pencil`. The object coordinates are fixed and known in advance. Therefore, visual recognition is responsible only for the semantic category, while the manipulation controller uses pre-calibrated world coordinates.

The final scene contains a square desktop, a mechArm 270 Pi with an adaptive gripper at the centre, six objects on a circular ring, and two open-top bins. Four positions contain tennis balls and two positions contain green wooden pencils. G6 was changed from a pencil to a tennis ball after the first batch test showed that the pencil at that position could not reach the required confidence consistently. This correction preserved the two-category task while improving the robustness of the complete arrangement.

Table 1: Basic requirements and completion evidence

Requirement	Implementation and evidence	Status
Simulated robotic arm and gripper	MechArm 270 Pi and adaptive gripper imported into Isaac Sim; ROS 2 trajectory controller connected through TCP.	Completed
Six objects and two bins	Four tennis balls and two pencils are arranged around the robot; the bins are open-top, deep, and physically collidable.	Completed
Circular layout	Object and bin centres use fixed coordinates around the robot-centred ring.	Completed
Virtual overhead camera	The final camera is 10 degrees tilted toward north, 1.10 m high, and aimed at the ring centre.	Completed
Category recognition	YOLO uses fixed image ROIs and six-frame majority voting; it does not calculate pickup coordinates.	Completed
Automatic sorting	The controller uses the locked categories to select the tennis or pencil bin and executes G1–G6 in order.	Completed
Error handling and records	Unknown classes stop the task safely; camera sweep reports, detections, parameters, and run recordings are retained.	Completed

2 Group Division of Work

The group work covered the complete simulation and validation chain rather than one isolated software module. The Experiment 3 workspace remains independent from Experiment 2. The group adapted the mechArm model and built the circular experiment scene, including the desktop, object ring, two deep bins, pencil geometry, collision bodies, and friction materials.

The main work packages were:

1. **Scene and physical modelling:** created the centred ring layout, open bins with collision walls, tennis balls, and a composite green wooden pencil with a cylindrical shaft, wood tip, and graphite tip. I added collision geometry for the pencil and tuned the object-gripper friction.
2. **Camera and perception:** connected the Isaac camera stream to ROS 2, reused the Experiment 1 YOLO model, synchronised fixed image ROIs with the tilted camera, and designed the six-frame majority-vote calibration process.
3. **Manipulation control:** retained the Experiment 2 mechArm parameters, numerical IK, smooth quintic interpolation, separate fast and slow motion speeds, lift clearance, direct release above the bin, and final HOME return.
4. **ROS 2 and Docker integration:** implemented the launch flow, TCP joint bridge, trajectory controller, virtual camera node, YOLO node, sorting state machine, recorder, and Docker build environment.
5. **Testing and evidence:** wrote contract and unit tests, performed the 13-view camera sweep, recorded detection metrics, preserved representative scene frames, and maintained the local GitHub repository for Experiment 3.

3 Principles of the Project Implementation

3.1 System Architecture

The system is divided into a Windows Isaac Sim process and a Dockerised ROS 2 process. Isaac Sim owns the scene, rigid-body dynamics, robot articulation, camera rendering, and TCP server. ROS 2 owns the visual classification, fixed-coordinate target generation, trajectory interpolation, task sequencing, and result recording.

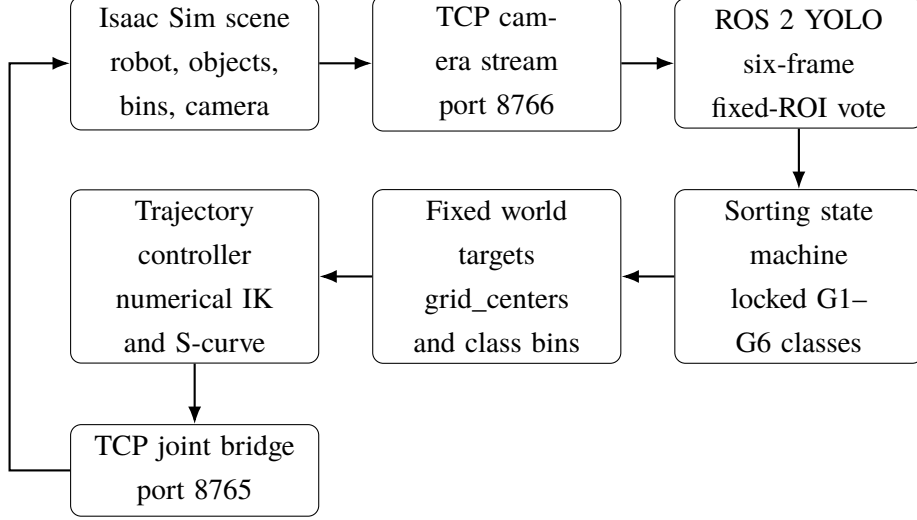


Figure 1: Data and control flow of the Experiment 3 simulation

3.2 Circular Layout and Fixed Coordinates

The robot is located at the origin of the desktop coordinate system. The six object centres and two bin centres are specified in `config/experiment3_sorting.yaml`. The controller uses these values directly. For a grid G_i , the pickup command is

$$\mathbf{p}_i = [x_i, y_i, z_i]^T = \text{grid_centers}[G_i]. \quad (1)$$

The class returned by YOLO selects the bin, but neither the YOLO image coordinate nor the detected box size changes $\mathbf{p}_i$.

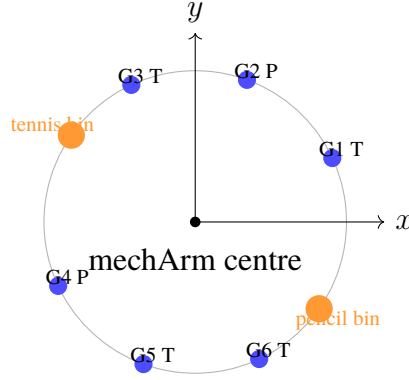


Figure 2: Schematic of the robot-centred circular arrangement; T denotes tennis ball and P denotes pencil.

3.3 Camera and Fixed-ROI YOLO Recognition

The production camera uses a resolution of 848×480 , focal length 18 mm, and height 1.10 m. A sweep tested 13 candidates: vertical, then 10, 15, and 20 degree tilts from four

azimuth directions. Each candidate produced five frames. The winning camera is positioned at approximately $(0, 0.19396, 1.10)$ m and looks at the origin.

The recognition pipeline is intentionally different from visual servoing. The six object locations are known. For each frame, the classifier crops the configured ROI for the requested grid and runs YOLO on that crop. After six frames, the node counts class votes. A class is accepted only if it is the unique winner with at least four votes. The sorting node receives six semantic decisions in one message, stores them, and then uses the fixed grid order for motion.

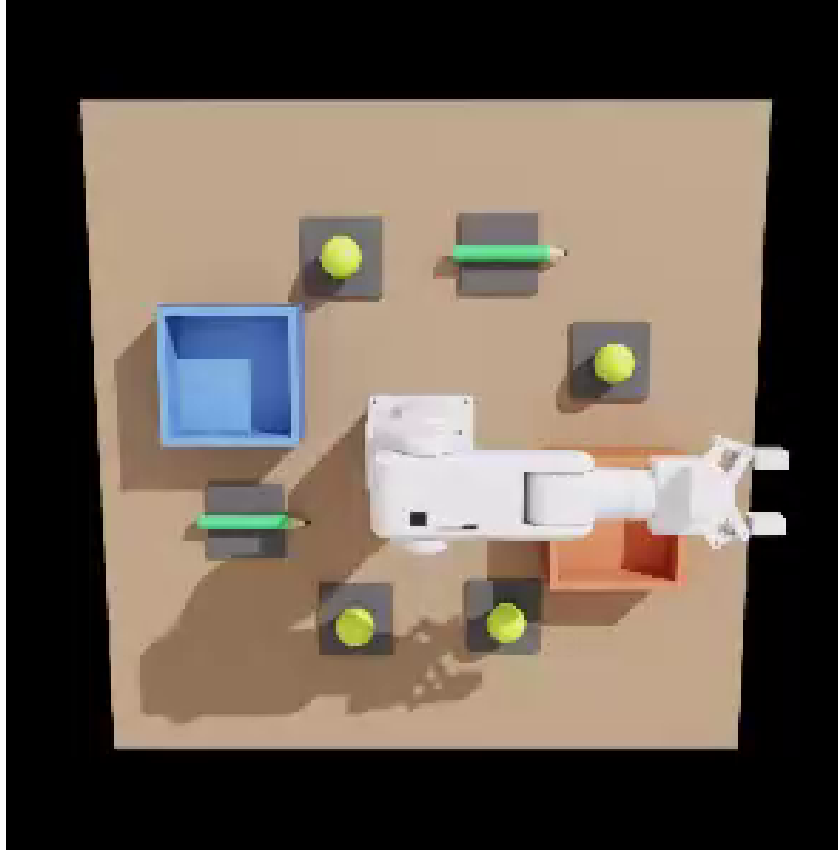


Figure 3: Final Experiment 3 scene after replacing G6 with a tennis ball. The robot is at the centre and the six objects and two bins surround it.

3.4 Manipulation and Motion Planning

The motion controller follows the stable strategy used in Experiment 2. The target pose is solved numerically while preserving the tool-axis constraint. Ordinary horizontal motion uses 20 deg/s and the final vertical approach, closure, lift, and release-related movement use 3 deg/s. A quintic smoothstep profile reduces acceleration discontinuities:

$$s(t) = 10\tau^3 - 15\tau^4 + 6\tau^5, \quad \tau = t/T, \quad 0 \leq \tau \leq 1. \quad (2)$$

For each object the arm moves to a pre-grasp height, descends slowly to the fixed grasp height, closes the gripper, lifts to $z = 0.133$ m, moves above the selected bin, opens the gripper to release the object, and proceeds directly to the next grid. The robot returns HOME only after G6.

4 Problems Encountered and Solutions

4.1 Camera Sweep Initially Produced Black Frames

The first sweep created 65 image files, but pixel inspection showed that all images had maximum intensity 0. Treating those files as valid evidence would have produced a false YOLO conclusion. The cause was that the sweep used an independent Replicator camera and advanced Replicator steps without following the normal Isaac World.step(render=True) loop. I changed the sweep to reuse the existing scene camera, initialise an Isaac World, play the world, and wait for rendered frames. I also added a non-black frame validator that aborts instead of recording invalid evidence. The second sweep produced 65 non-black images.

4.2 Full-Image YOLO Was the Wrong Evaluation Boundary

When the model was run on the complete 848×480 image, it detected tennis balls but returned no pencils. This initially suggested that camera angle alone could not solve the problem. I compared the classifier implementation and found that the production node already cropped a fixed ROI before YOLO inference. Re-evaluating each known grid ROI reproduced the actual runtime boundary and recovered pencil detections. This changed the sweep evaluator from full-image detection counting to projected fixed-ROI classification. The result was more meaningful and directly connected to the real sorting pipeline.

4.3 The 90-Degree View Was Not Uniformly Robust

The original vertical view was visually clear, but two pencil positions had weak confidence. I tested 13 views with the same model and threshold. The 10-degree north view achieved 6/6 correct decisions in all five frames and raised the weak pencil confidences. The camera pose and all world pickup coordinates were then synchronised with the selected view. This was a useful example of separating geometric calibration from learned detection: the camera changed, but the robot did not chase image coordinates.

4.4 Batch Recognition Revealed an Unstable G6 Pencil

After changing the controller to classify all six grids before motion, the first runtime batch produced stable results for G1–G5 but marked G6 unknown: its six votes did not reach the required four-vote threshold. I intentionally kept the safety stop instead of forcing a class or grabbing blindly. Since the experiment only requires the two categories and G6 is a fixed design position, I replaced the unstable G6 pencil with a tennis ball, regenerated the USD scene, and updated the YAML object record, ROI shape, and colour metadata. A later runtime capture then showed G6 classified as a tennis ball with 6/6 votes.

4.5 Middleware, Packaging, and Numerical Issues

Several lower-level issues also required engineering decisions:

- A Docker build failed because a ROS package download returned HTTP 502. Retrying the package installation resolved the transient registry failure.
- The Python environment initially contained NumPy 2.x while the ROS cv_bridge binary had been compiled against NumPy 1.x. Downgrading to NumPy 1.21.5 removed the binary API warning and restored cv_bridge compatibility.
- PowerShell quoting caused false syntax errors when Python code contained nested quotes. I changed the verification commands to use the container’s direct Python entry point and simpler quoting.
- Isaac reported unresolved visual references and PhysX iteration warnings. They did not stop the TCP state stream; I recorded them as non-blocking warnings and kept the robot and camera data paths under observation.
- One early execution stopped at numerical IK because the requested pose was near the orientation boundary. The final controller preserves the Experiment 2 arm parameters and uses safe pre-grasp/lift heights rather than relaxing the physical constraints blindly.

These problems shaped my implementation choices. I used evidence gates for images, fixed coordinate contracts for grasping, explicit safe-stop behaviour for uncertain perception, and repeated static tests after each structural change. This made the system easier to explain and reduced the risk of a visually plausible but physically incorrect result.

5 Verification and Results

5.1 Camera View Sweep

The final sweep used the Experiment 1 model `pencil_tennis_yolo26n_best.pt`, confidence threshold 0.05, six fixed ROIs, and 65 valid frames. The selected 10-degree north view achieved the following verified result:

Table 2: Camera sweep result

Metric	Result
Candidate views	13
Frames per view	5
Total valid frames	65
Best view	10-degree north tilt
Minimum correct objects per frame	6/6
Pencil recall	1.000
Tennis-ball recall	1.000
Mean confidence	0.836

The report directory contains the full ranking CSV, per-grid detections, camera recommendation YAML, model checksum, application record, and representative annotated frames. The camera sweep evidence is therefore reproducible rather than based on a single favourable image.

5.2 Batch Recognition and Runtime Evidence

The first batch runtime after implementation produced G1 tennis ball, G2 pencil, G3 tennis ball, G4 pencil, G5 tennis ball, and G6 unknown. The resulting safe stop demonstrated that the 4/6 rule was active. After replacing G6 with a tennis ball, a subsequent recorded run showed G1–G6 all classified with 6/6 votes, including G6 as `tennis_ball`. The run recorder’s summary CSV was empty because the capture ended during execution, so I do not claim a completed six-cycle success count from that file. The video and terminal evidence do verify the batch request, six grid classifications, and the beginning of the fixed-coordinate pick sequence.

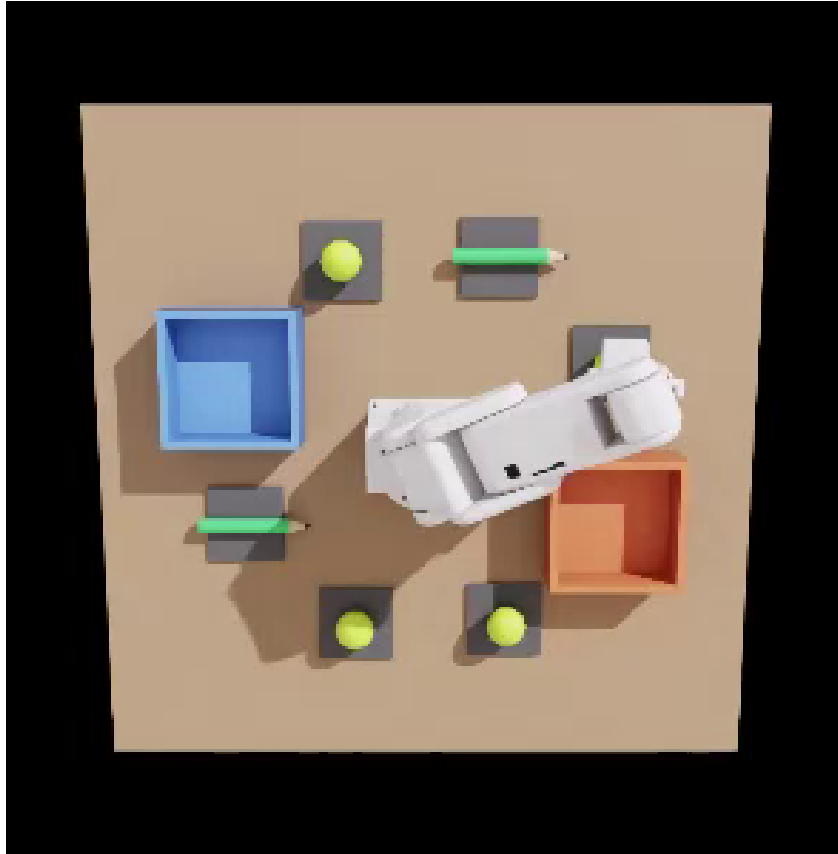


Figure 4: Representative runtime frame while the robot executes the fixed-coordinate manipulation sequence.

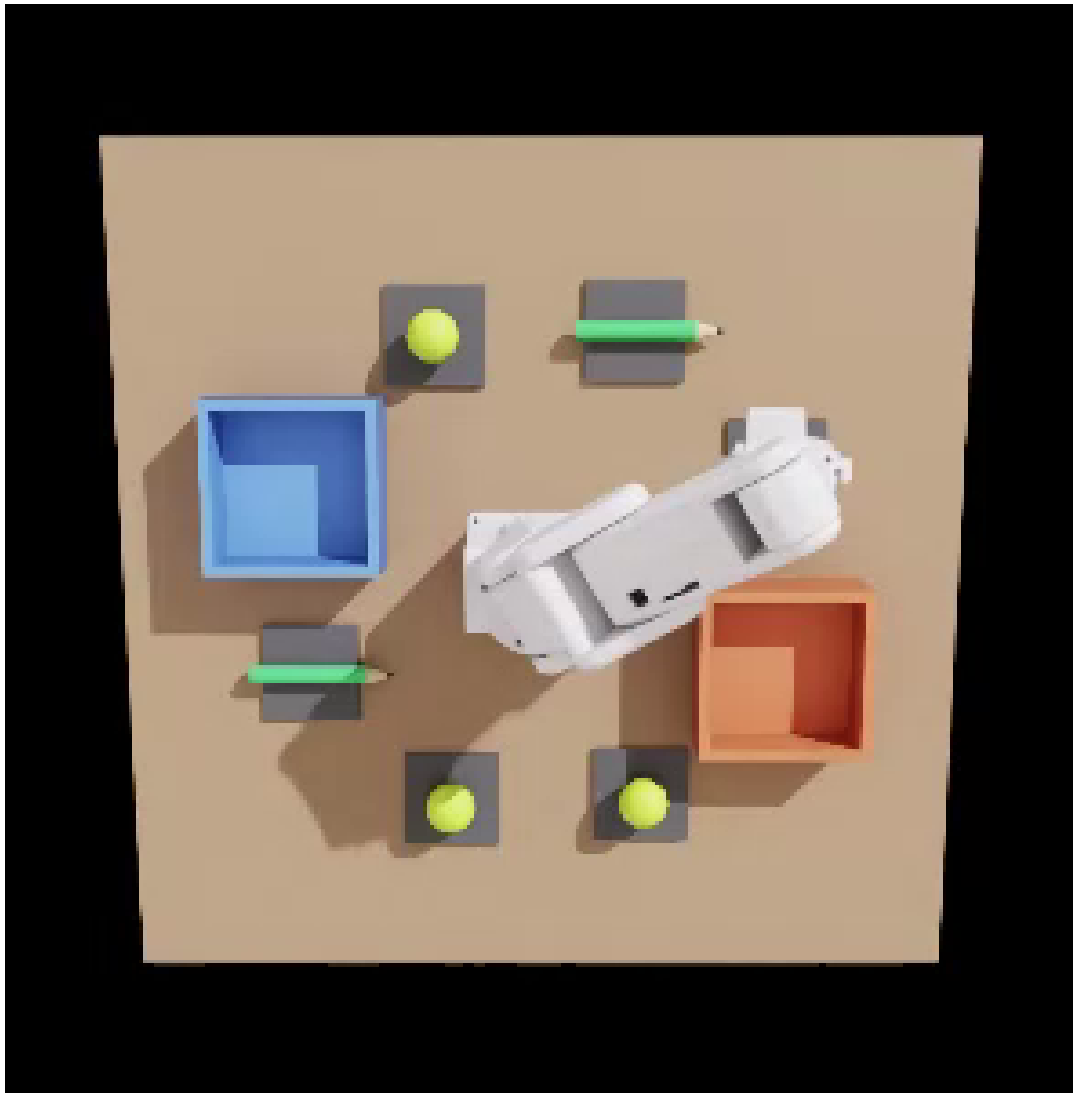


Figure 5: Another runtime scene frame showing the arm moving between a fixed object position and the selected bin.

```
[pick_place_recorder_node-6] [INFO] [1788971734.026090802] [pick_place_recorder_node]: recording experiment results in /workspace/mecharm_exp3/results/experiment3/20260909-163533
[isaac_trajectory_controller-2] [INFO] [1788971734.040372098] [isaac_trajectory_controller]: trajectory action ready: /mecharm_controller/follow_joint_trajectory
[virtual_camera_node-3] [INFO] [1788971734.292342595] [virtual_camera_node]: connected to Isaac camera stream 127.0.0.1:8766
[yolo_classifier_node-4] [INFO] [1788971735.958275371] [yolo_classifier_node]: loading fixed-ROI YOLO model: /workspace/mecharm_exp3/models/pencil_tennis_yolo26n_best.pt
[isaac_tcp_bridge-1] [INFO] [1788971736.718336257] [isaac_tcp_bridge]: received 612 raw bytes from Isaac TCP bridge
[isaac_tcp_bridge-1] [INFO] [1788971736.735128445] [isaac_tcp_bridge]: received Isaac TCP state with 12 joints
[isaac_tcp_bridge-1] [INFO] [1788971736.735594028] [isaac_tcp_bridge]: publishing /joint_states with 12 joints
[sorting_task_node-5] [INFO] [1788971737.001032931] [sorting_task_node]: GRIPPER_WAIT_BEFORE INITIAL_GRIPPER_OPEN 1.00s
[sorting_task_node-5] [INFO] [1788971738.054541645] [sorting_task_node]: DIRECT_STAGE INITIAL_GRIPPER_OPEN duration=1.26s speed=0.120rad/s
[sorting_task_node-5] [INFO] [1788971739.383707686] [sorting_task_node]: GRIPPER_WAIT_AFTER INITIAL_GRIPPER_OPEN 1.50s
[sorting_task_node-5] [INFO] [1788971740.927324491] [sorting_task_node]: DIRECT_STAGE HOME duration=4.40s speed=20.0deg/s
[sorting_task_node-5] [INFO] [1788971745.747325067] [sorting_task_node]: VISION_BATCH_REQUEST: classifying all fixed grids over 6 frames
[yolo_classifier_node-4] [INFO] [1788971747.616284389] [yolo_classifier_node]: YOLO BATCH grid=G1 class=tennis_ball votes=6/6 confidence=0.964
[yolo_classifier_node-4] [INFO] [1788971747.645639840] [yolo_classifier_node]: YOLO BATCH grid=G2 class=pencil votes=6/6 confidence=0.075
[yolo_classifier_node-4] [INFO] [1788971747.675446110] [yolo_classifier_node]: YOLO BATCH grid=G3 class=tennis_ball votes=6/6 confidence=0.954
[yolo_classifier_node-4] [INFO] [1788971747.705410190] [yolo_classifier_node]: YOLO BATCH grid=G4 class=pencil votes=6/6 confidence=0.329
[yolo_classifier_node-4] [INFO] [1788971747.735507929] [yolo_classifier_node]: YOLO BATCH grid=G5 class=tennis_ball votes=6/6 confidence=0.752
[yolo_classifier_node-4] [INFO] [1788971747.768764549] [yolo_classifier_node]: YOLO BATCH grid=G6 class=tennis_ball votes=6/6 confidence=0.966
[sorting_task_node-5] [INFO] [1788971747.833519109] [sorting_task_node]: VISION_BATCH_READY: locked classes for 6 fixed grids
[sorting_task_node-5] [INFO] [1788971747.879632266] [sorting_task_node]: PICK: fixed G1: YOLO classified tennis_ball
[sorting_task_node-5] [INFO] [1788971747.961483100] [sorting_task_node]: NUMERIC_IK_REQUEST PICK_PREGRASP xyz=(0.180,0.080,0.125)
[sorting_task_node-5] [INFO] [1788971748.048769442] [sorting_task_node]: NUMERIC_IK_SUCCESS PICK_PREGRASP position_error=0.0036 axis_dot=0.9872
[sorting_task_node-5] [INFO] [1788971748.094352905] [sorting_task_node]: DIRECT_STAGE PICK_PREGRASP du
```

Figure 6: Terminal evidence of one batch request, six fixed-grid classifications, and the first fixed-coordinate pick. In particular, G6 is classified as a tennis ball with six of six votes.

5.3 Software Verification

The final local test command ran both the Experiment 3 tests and the ROS package tests:

Listing 1: Final static verification command

```
$env:PYTHONPATH='<experiment3-root>/src/mecharm_pick_place'
python -m pytest <experiment3-root>/tests <experiment3-root>/src/
    mecharm_pick_place/test -q
```

It returned 78 passed tests. Python compilation also passed for the modified Isaac and ROS 2 modules, and the Docker colcon build completed for both `mecharm_pick_place` and `mycobot_description`.

5.4 GitHub Version-Control Evidence

The Experiment 3 source, configuration, model, USD scene, runtime evidence, and report materials are maintained in the public repository <https://github.com/xhr-CHN/mecharm-experiment3-sorting>. The commit history records the work in incremental stages rather than as one undifferentiated upload: documentation, runtime configuration, Isaac assets, ROS 2 code, validation tools, YOLO/runtime evidence, the G6 scene correction, and the individual report are all separately identifiable. Figure 7 provides a direct screenshot of the repository’s `main` branch commit page.

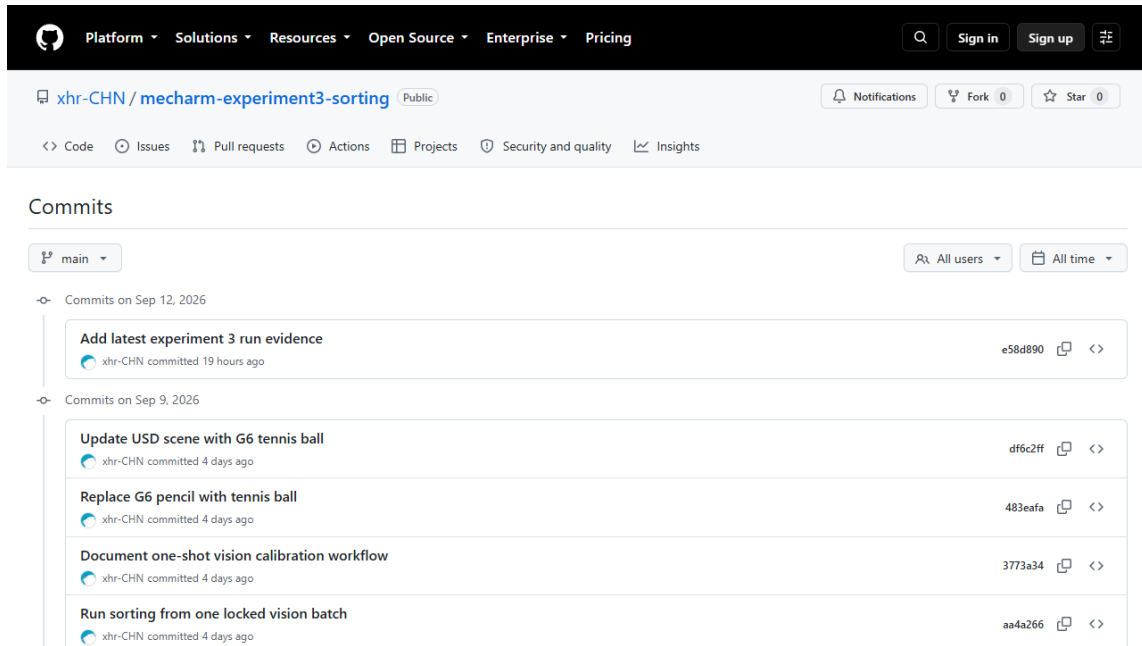


Figure 7: GitHub commit history for the Experiment 3 repository, showing the incremental project records.

6 Real-Robot Validation

The real-robot half of the group experiment uses a RoboMaster EP rather than the six-axis mechArm simulation. The real program calls the RoboMaster Python SDK directly and does not depend on ROS 2. The EP carries a camera and uses a fixed six-grid floor layout. The YOLO model contains three classes: cup, mouse, and bottle. The program executes only cup and bottle candidates; a mouse is recorded as an unsupported object.

6.1 Real System and Visual Decision Rules

The camera observes six fixed floor regions after the arm returns to a repeatable centre pose. Each scan uses 12 frames. A candidate needs at least five hit frames and a vote share of at

least 0.70. The detection-box centre maps the candidate to G1–G6. A shape guard adds a practical protection for the current metal cup: aspect ratios at or below 1.75 are treated as cup, ratios at or above 2.00 as bottle, and the interval between them is rejected as ambiguous. This rule is a fixed-scene safety measure, not a claim that the model has been retrained.

The execution loop is:

recenter $\rightarrow$ scan $\rightarrow$ select G1–G6 $\rightarrow$ route $\rightarrow$ grasp $\rightarrow$ return $\rightarrow$ place $\rightarrow$ rescan. (3)

The chassis uses measured field corrections of ± 0.27 m for horizontal alignment, 0.06 m for the near row, and 0.48 m for the far row. These are open-loop values and remain sensitive to wheel slip, floor material, battery level, and load. Far-row access is protected by a lane interlock: G4, G5, and G6 are allowed only after the corresponding near-row channel is clear.

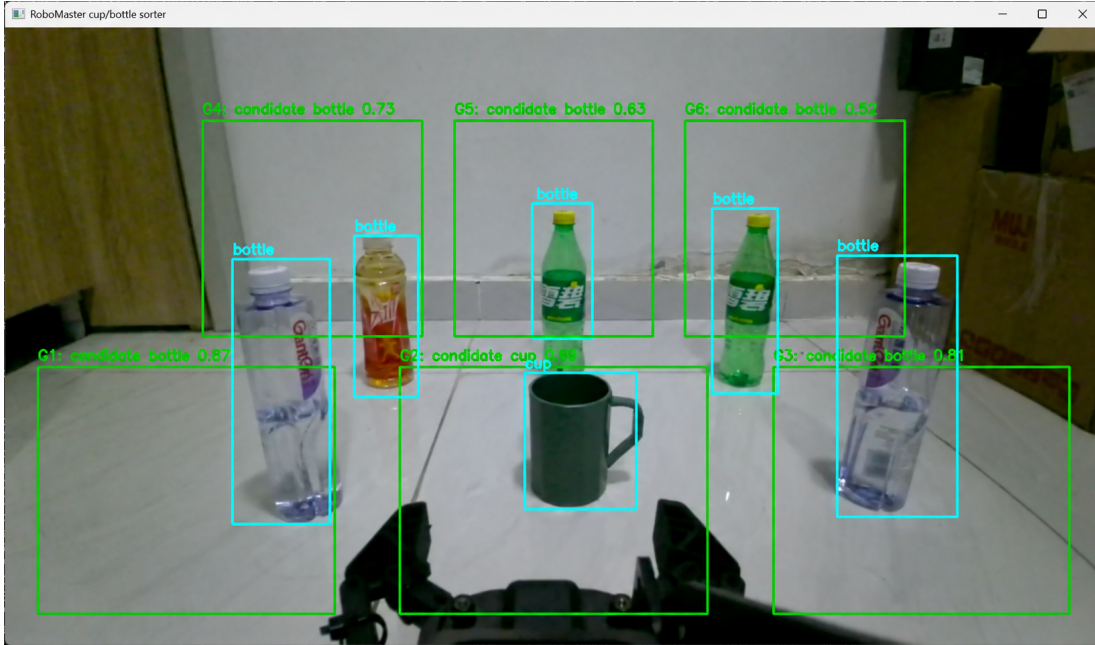
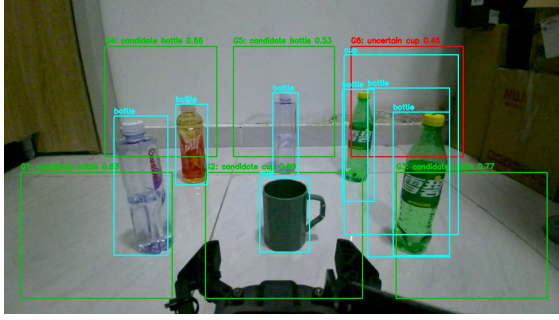


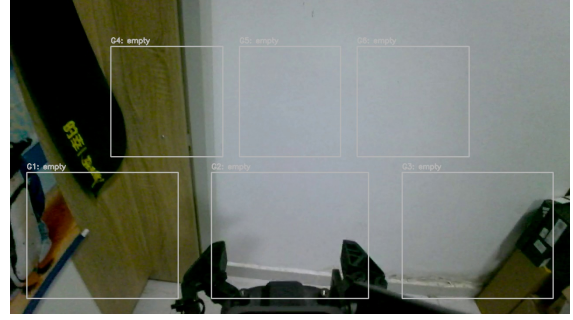
Figure 8: Real-camera six-grid detection evidence. The image shows fixed grid regions, YOLO boxes, class labels, confidence values, and the camera mounted on the RoboMaster platform.

6.2 Real Pick-and-Place Motion

The real arm uses class-specific nominal points. The cup grasp contact point is $[205, -60]$ mm and the bottle grasp contact point is $[205, -15]$ mm. The arm first moves down, extends horizontally, closes the gripper, lifts, and returns to the centre. The robot then rotates toward the corresponding classification area, releases the object, and returns to the scan position. The points are empirical values for the current robot, camera, floor, and object geometry rather than universal coordinates.



(a) Initial scan and candidate selection



(b) Later scan after a sorting action

Figure 9: Representative frames from the real RoboMaster EP run.

6.3 Real Validation Result and Evidence Boundary

The representative run `run_20260913_012927_039424` used execute mode, completed four scans, and recorded two source-grid-clearing actions: one bottle and one cup. Its stop reason was the configured two-object limit. The package contains the corresponding event log, task summary, scan images, and two versions of the video. The raw video is approximately 84.43 s at 30 FPS; the PPT embedded version is approximately 56.29 s at 45 FPS. The source package also reports 88 offline unit tests passed across the final program and staged test variants.

Table 3: Representative real-robot validation record

Item	Verified value
Robot platform	RoboMaster EP
Representative run	20260913_012927_039424
Execution mode	execute=true
Completed source-grid clears	2
Classes in completed actions	bottle=1, cup=1
Number of scans	4
Offline unit tests	88 passed

The two-object real video demonstrates the integrated camera, category decision, chasis route, arm grasp, release, and rescan loop. It does not constitute a six-object real-robot acceptance test, and the source-grid-cleared count alone does not prove that every object entered the correct classification tray. This boundary is kept explicit so the simulation and real evidence remain technically comparable without overstating the result.

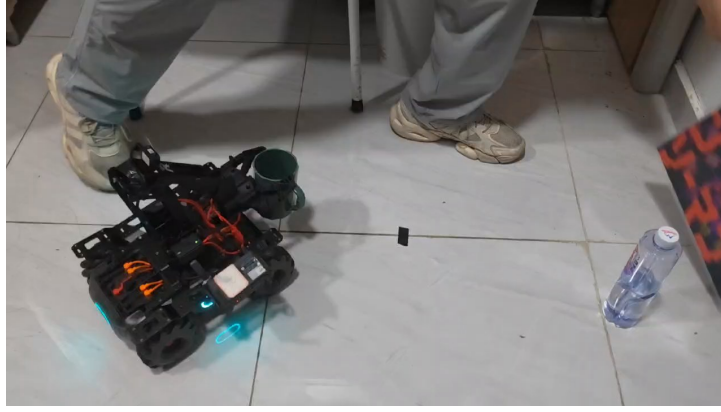


Figure 10: Cover frame for the preserved two-object RoboMaster EP demonstration video.

7 Conclusion

The group completed an independent Experiment 3 package combining a six-object mechArm 270 Pi simulation with a separately documented RoboMaster EP real-robot validation. The simulation combines a robot-centred ring scene, open deep bins, tennis-ball and green-pencil models, a calibrated tilted virtual camera, fixed-ROI YOLO classification, six-frame majority voting, fixed-world-coordinate manipulation, numerical IK, smooth motion, safe-stop handling, and final HOME return. The real stage demonstrates the same task idea through the EP’s camera, fixed six-grid mapping, shape guard, lane interlock, class-specific arm points, and two-object video evidence.

The most important design decision was to keep perception and manipulation responsibilities separate. YOLO answers “what category is at this known grid?”; it does not answer “where should the robot grasp?”. This makes the system more stable, easier to test, and consistent with the fixed-coordinate experimental requirement. The 13-view sweep achieved 6/6 classification in every tested frame for the selected view, and 78 static tests passed after the final scene correction.

The key limitation is the platform and task boundary. The real evidence covers two cup/bottle actions on a RoboMaster EP, not six tennis-ball/pencil actions on the mechArm. The simulation also lacks a complete machine-readable six-placement success table in the latest run. A stronger final acceptance package should preserve one result record for every placement, repeat the six-object run under one fixed condition, and add calibrated physical feedback for grasp confirmation.

References

1. NVIDIA, “Isaac Sim Documentation, Version 5.1.0,” <https://docs.isaacsim.omniverse.nvidia.com/5.1.0/>.
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